Signal & Data Analysis in Neuroscience 2020

Independent Component Analysis (ICA)

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Overview

- Moments
- ICA

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Moments - reminder

- The n^{th} moment

$$\mu'_n = \int_{-\infty}^{\infty} (x-c)^n f(x)dx$$

- The n^{th} **central** *moment*

$$\mu_n = E((X-\mu)^n)$$

- The n^{th} **normalized** moment

$$\mu_n = \frac{E((X-\mu)^n)}{\sigma^n}$$

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Describing the distribution

- The first four moments are useful for describing aspects of the distribution

1. 1st moment - **Mean** (**central value**)
2. 2nd moment - **Variance** (**dispersion**)
3. 3rd moment - **Skewness** (**asymmetry**)
4. 4th moment - **Kurtosis** (**peakedness**)

(Many of the next slides based on: Jingfeng Xiao, UNC)

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Skewness I

- **Skewness** (3rd normalized moment) measures the degree of asymmetry exhibited by the data

$$skewness = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3}{n\sigma^3}$$

- **Skewness** = 0 → The histogram is **symmetric** about the mean
- **Positive** skewness → large positive values
- **Negative** skewness → large negative values

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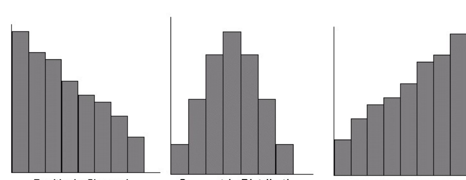
Skewness II



Positively Skewed Histogram

Symmetric Distribution Histogram

Negatively Skewed Histogram

(http://library.thinkquest.org/10030/3smodsas.htm)

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Skewness III

- **Positive skewness**
  - More observations below the mean than above it
  - The mean is greater than the median
- **Negative skewness**
  - A small number of low observations and a large number of high ones
  - The median is greater than the mean

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Kurtosis I

- **Kurtosis** measures how peaked the histogram is

$$kurtosis = \frac{\sum_{i}^{n}(x_i - \overline{x})^4}{n\sigma^4} - 3$$

- The **kurtosis** (standardized 4th moment) of a **normal distribution** is 0
- **Kurtosis** characterizes the relative **peakedness** or **flatness** of a distribution compared to the normal distribution

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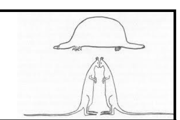
Kurtosis II



- **Platykurtic (platy = broad)** – **kurtosis < 0**, the frequencies throughout the curve are closer to be equal (i.e., the curve is more **flat** and **wide**)
- **Leptokurtic (lepto = slender)** – **kurtosis > 0**, there are high frequencies in only a small part of the curve (i.e, the curve is more **peaked**)

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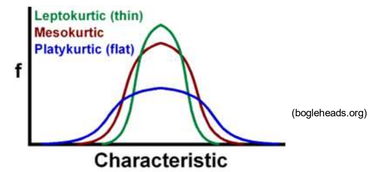
Kurtosis III



- **Kurtosis** is based on the size of a distribution's tails.
- **Negative** kurtosis (**platykurtic**) – distributions with short tails
- **Positive** kurtosis (**leptokurtic**) – distributions with relatively long tails

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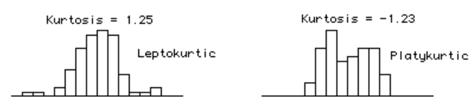
Kurtosis IV



- These two distributions have the same **variance**, approximately the same **skew**, but **differ** markedly in **kurtosis**.

Source: http://davidmlane.com/hyperstat/A53638.html

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Overview

- Moments
- ICA

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Independent component analysis

- Independent component analysis (ICA) is a computational method for separating a multivariate signal into additive subcomponents supposing the mutual statistical independence of the non-Gaussian source signals. (Wikipedia)
- ICA separates the data into statistically independent components as opposed to achieving second order decorrelation by PCA.

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In this presentation I use slides from: Jochen Triesch

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Uncorrelated vs. Independent

- PCA leads to uncorrelated variables

$E(y_1 \cdot y_2)=E(y_1) \cdot E(y_2)$

- ICA leads to independent variables.

For every pair of functions f_1 & f_2:
$E(f_1(y_1) \cdot f_2(y_2))=E(f_1(y_1)) \cdot E(f_2(y_2))$

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Uncorrelated vs. Independent example

For joint values of (y_1,y_2) → (0,1),(0,-1),(1,0),(-1,0) at p=0.25

- The variables y_1 & y_2 are uncorrelated:

$E(y_1) \cdot E(y_2) = 0 = E(y_1 \cdot y_2)$

- The variables y_1 & y_2 are not independent:

For the function $f_1(x)=f_2(x)=x^2$

$E(y_1^2 \cdot y_2^2) = 0$ while $E(y_1^2) \cdot E(y_2^2) = 0.25$

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Independent Component Analysis

- There are many algorithms for performing ICA many of which rely on information theory. For example: maximization of the output's joint entropy while minimizing the mutual information between the different channels.
- Unlike PCA, there is no closed solution for discovering the independent components.
- Transform X → Y such that:

$I(y_i;\, y_j)$ is minimal and $I(X, Y)$ is maximal

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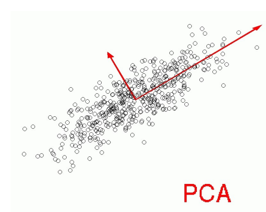
ICA vs. PCA

- Principal Component Analysis (PCA) finds directions of maximal variance in Gaussian data (second-order statistics).



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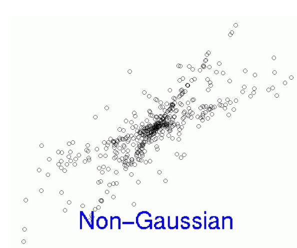
Non Gaussian data

- Non Gaussian data has higher order statistical dependencies between variables.



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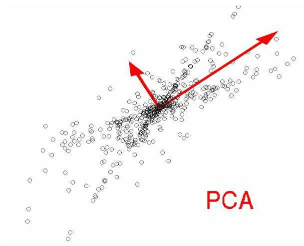
Non Gaussian data - PCA



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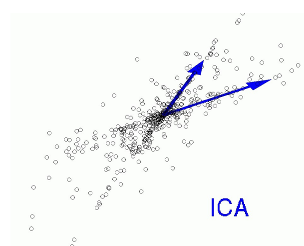
Non Gaussian data - ICA

ICA finds directions of maximal independence in non-Gaussian data (higher-order statistics).



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Blind source separation (BSS)

- Blind signal separation, also known as blind source separation, is the separation of a set of signals from a set of mixed signals, without the aid of information about the source signals or the mixing process. (Wikipedia)

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Example: Cocktail party problem

- Many speakers as sound sources and array of microphones, where each microphone is picking up a different mixture of the speakers.
- Can we separate the individual speakers given just the microphone signals, i.e. without knowing how the speakers' signals got mixed?

(Examples: http://www.cnl.salk.edu/~tewon/Blind/blind_audio.html)

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BSS & ICA

- BSS may be performed by ICA or other methods
- ICA may be used to solve problems other than BSS.
- However, in many cases ICA solution is described as a subset of the BSS problem.

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Mathematical formulation

- The simplest and most restrictive definition:

1. *n* Sources are independent and non-gaussian: $s_1, \ldots, s_n$
2. Sources have zero mean
3. *n* observations are linear mixtures: $\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t)$

- Assuming the inverse of ***A*** exists: $\mathbf{W} = \mathbf{A}^{-1}$
- ICA estimates ***W***: $\hat{\mathbf{s}}(t) = \hat{\mathbf{W}}\mathbf{x}(t)$
- Leading back to the sources: $\mathbf{y}(t) = \hat{\mathbf{W}}\mathbf{x}(t)$

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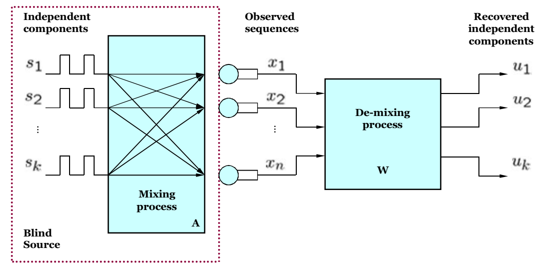
ICA graphically



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Restrictions of ICA

- Requirements:
  1. Sources are independent
  2. Sources are non-gaussian (at most one can be gaussian)
- Ambiguities of the solution:
  1. Sources can be estimated only up to a constant scale factor
  2. Sources may not be recovered in any order

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ICA the catch…

- Unlike PCA which has a closed solution, ICA has multiple algorithms for optimizing the approximation of W. It is thus also a very active field of research.
- Multiple methods:
  - Maximizing non-gaussianity of Y
  - Minimizing the mutual information of Y
  - Maximum likelihood estimation (may utilize prior knowledge about the distributions)

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Maximization of non gaussianity

- The Central Limit Theorem
  Distribution of a sum of independent random variables tends toward a Gaussian distribution
- Measures of non-gaussianity
  1. **Kurtosis**:
  Zero for a Gaussian variable, and greater than zero for most non-Gaussian random variables.
  2. **Negentropy**: $J(y) = H(y_{gauss}) - H(y)$
  A Gaussian variable has the largest entropy among all random variables of equal variance .
  Thus, negentorpy is always non-negative, and it's zero *iff* y has a Gaussian distribution.

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Minimization of Mutual Information

- Using the concept of differential entropy, mutual information I between *m* random variables can be define as following,

$$Y = (y_1, y_2, ..., y_m) \qquad vector$$

$$I(Y) = \sum_{i=1}^{m} H(y_i) - H(Y)$$

- Mutual information is a natural measure of the dependence between random variables.
- Minimizing the mutual information can recover individual independent components.
- Equivalent to minimizing the negentropy.

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Maximum Likelihood Estimation

- Essentially equivalent to minimizing mutual information $\hat{W} = \arg\max_W \prod_{i=1}^{n} p(x_i | W)$
- Log likelihood is defined as following,

$$\hat{W} = \arg\max_W \sum_{i=1}^{n} \log[p(x_i | W)]$$

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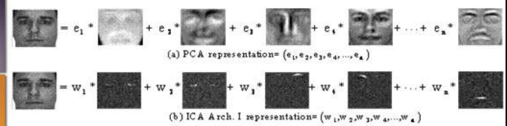
Example - Faces



(http://annenberg.edu)

- Global features extracted using PCA (top).
- Local features extracted using ICA (bottom).

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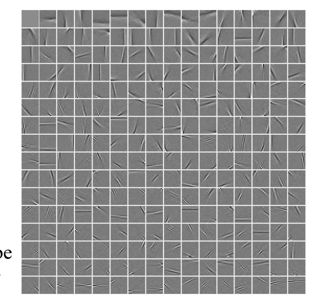
Examples – Natural scenes

- What are the "independent sources" of natural scenes?
- They look similar to V1 simple cells: localized, oriented, bandpass
- Finding ICs may be principle of sensory coding in cortex!



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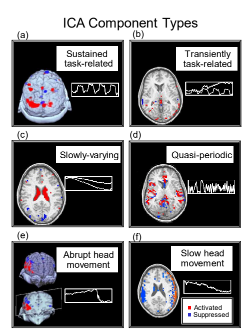
Example - Functional Brain Imaging

- Functional magnetic resonance imaging (fMRI) data are noisy and complex.
- ICA identifies concurrent hemodynamic processes.
- Does not require *a priori* knowledge of time courses or spatial distributions.



(Slide Dr. M. McKeown)

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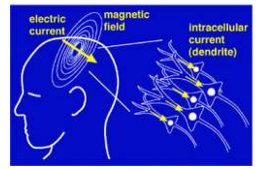
Example - MEG

- Separation of Artifacts in MEG. MEG measures the neuron electrical signal with a good spatial and temporal resolution. Assuming that brain activities are independent from the source of artifacts like eye movement or blinks (anatomically and physiologically), we can use ICA to analyze the interested part of signal.



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ICA - Other links

- ICA demo step-by-step
  - http://www.cis.hut.fi/projects/ica/icademo/
- Aapo Hyvarinen
  - http://www.cis.hut.fi/aapo/papers/NCS99web/node11.html

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